

EXPERIENCE CHANNELS - ROLE AND ORGANISATION CONTEXT

- S** App Shell + MFE
shipper + transporter workspaces
- O** Operator + tenant admin
control tower, finance, policy
- M** Driver mobile
tasks, tracking + evidence
- A** Partner API consumers
integrations + report extraction

EXTERNAL DEPENDENCIES - CONTRACTS

- I** Identity federation
WorkOS + enterprise IdPs
- E** Operational + enterprise
telematics, ERP, maps + payments
notifications + regulatory

NATS TRUST - OPERATOR TO CREDENTIALS

- O** Operator
trust root
- T** Tenant account
one per market
- R** Org roles
RBAC + ABAC
- C** Short-lived creds
users + workloads

APPLICATION AND INTEGRATION BOUNDARY - POLICY-ENFORCED INGRESS

- E** Regional edge + WAF
TLS, routing, rate limits
country endpoint policy
- I** Identity + policy broker
membership, RBAC + ABAC
tenant + org context
- B** BFF + API facade
role-shaped commands + reads
no direct database access
- R** Reporting + data API
scoped extracts + async export
masking + audit
- A** Integration adapters
webhooks, polling + events
anti-corruption boundary

CONTROL PLANE - LINEBOOKER TECH PLATFORM AUTHORITY

- T** Tenant + organisation
lifecycle + membership
authority boundaries
- I** Identity + membership
roles + policy attributes
Linebooker authorization
- N** NATS provisioning
accounts + credentials
claims + placement
- R** Reference + config
country + market policy
currency defaults
- E** Entitlements + admin
plans, billing + support
operator controls
- O** Observe + audit
health + immutable record
masked regional telemetry

DOMAIN SERVICE PLANE - INDEPENDENTLY DEPLOYABLE BUSINESS OWNERS

- M** Marketplace + tender
demand, bids, awards
owns tender state
- L** Loads + trips + routes
allocation + execution
owns movement state
- F** Fleet + tracking
assets, drivers, ETA
owns capacity + position
- D** Documents
evidence + compliance
owns metadata + policy
- D** Disputes + acceptance
proof gates + exceptions
owns dispute state
- B** Billing + settlement
rating, invoice, payment
owns commercial state

PLATFORM FOUNDATIONS - MESSAGING, WORKFLOW AND DISTINCT DATA RESPONSIBILITIES

- N** NATS Core + JetStream
RPC, commands + durable facts
tenant account boundary
- W** Temporal workflows
long-running coordination
services retain business rules
- P** Service-owned Postgres
transactional source of truth
database per service owner
- Q** Projections + NATS KV
read models + fast lookup
derived, rebuildable state
- S** Document store adapter
S3-compatible evidence store
large files outside NATS

Security, residency + immutable audit

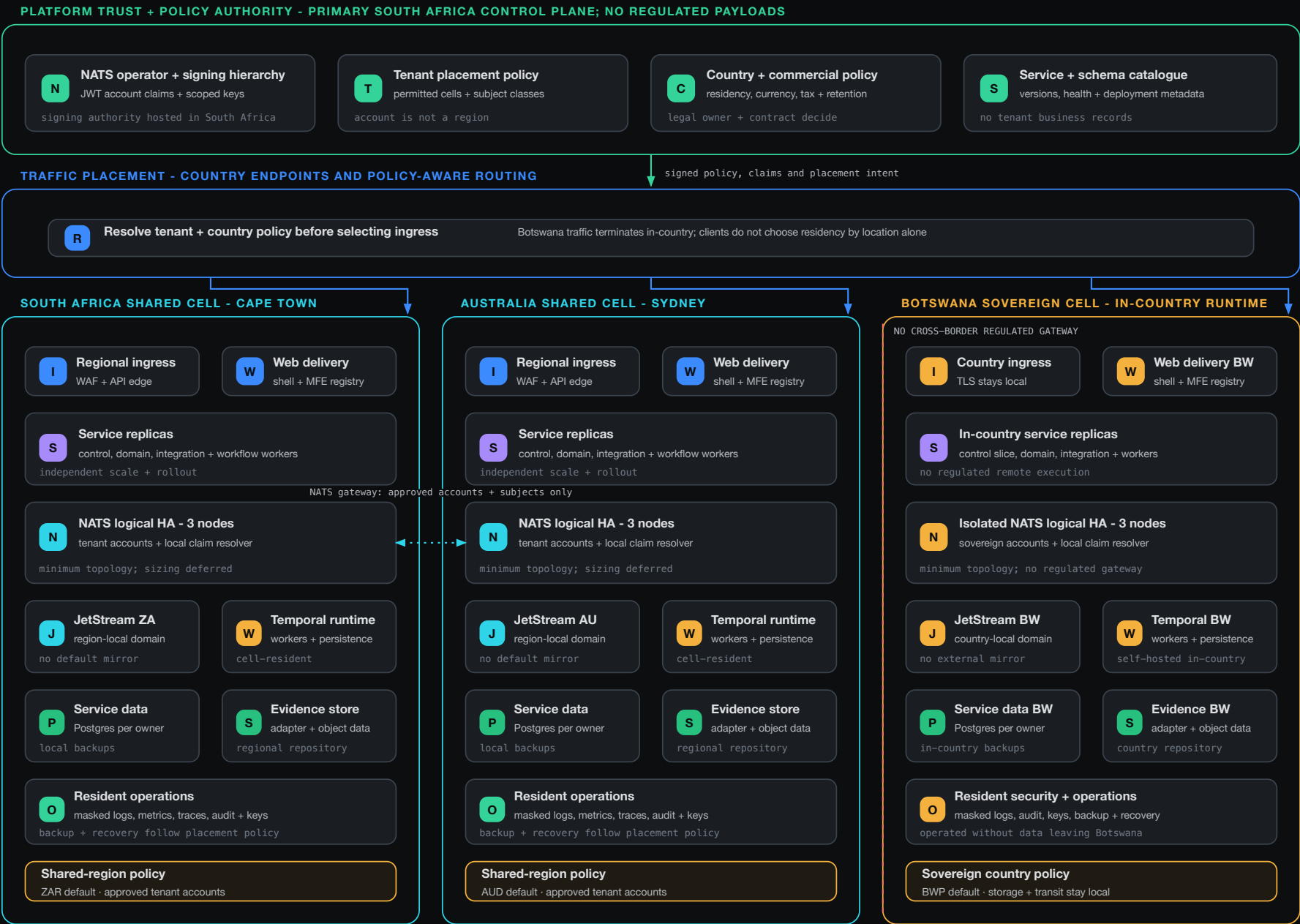
- S** Security, residency + immutable audit
tenant separation · RBAC/ABAC · TLS · encryption at rest · masked logs · country policy
durable business history without requiring universal event-sourced reconstruction

ACRONYMS

ABAC Attribute-based access control	POD Proof of delivery (not a Kubernetes pod)
API Application programming interface	RBAC Role-based access control
BFF Backend for frontend	RPC Remote procedure call
ERP Enterprise resource planning	S3 The object-storage API; S3-compatible means any vendor implementing it
ETA Estimated time of arrival	TLS Transport Layer Security
IdP Identity provider	WAF Web application firewall
KV Key/value store (NATS KV)	WorkOS A commercial identity provider, used as an example
MFE Micro-frontend	
NATS The messaging system; a product name, not an abbreviation	

Regional Deployment Model

This view applies the logical platform to three initial placement cells. South Africa and Australia may exchange explicitly approved NATS subject classes through gateways; Botswana is a sovereign in-country cell whose regulated business data, transit, operations and recovery do not cross the national boundary.



I **Country ingress**
TLS stays local

W **Web delivery BW**
shell + MFE registry

S **In-country service replicas**
control slice, domain, integration + workers
no regulated remote execution**N** **Isolated NATS logical HA - 3 nodes**
sovereign accounts + local claim resolver
minimum topology; no regulated gateway

J **JetStream BW**
country-local domain
no external mirror

W **Temporal BW**
workers + persistence
self-hosted in-country

P **Service data BW**
Postgres per owner
in-country backups

S **Evidence BW**
adapter + object data
country repository

O **Resident security + operations**
masked logs, audit, keys, backup + recovery
operated without data leaving Botswana**Sovereign country policy**
BWP default · storage + transit stay local

Proposed initial topology. Use three NATS nodes per cell as the logical high-availability minimum and one local JetStream domain; capacity remains an L3/L4 decision. Permit gateway routing only between shared cells and explicitly approved tenant accounts and subjects; do not mirror JetStream across regions by default. Botswana's sovereign border rejects every regulated cross-border gateway path.

TOPOLOGY	Start with independent regional clusters. Gateways are an explicit policy exception for shared cells, not an assumed global mesh.
NATS	The trust authority has a home. Operator signing keys and the primary catalogue run in South Africa; each cell uses local resolvers and short-lived credentials.
TENANT	Account and region are separate axes. One tenant account may exist only in its approved cells; organisation roles do not create new NATS accounts.
JS	JetStream is local by default. Streams, KV and selected durable facts remain in their assigned domain unless an approved design adds replication.
BOTSWANA	The sovereign boundary includes transit. Compute, NATS traffic, workflow state, databases, documents, telemetry, keys, backups and recovery stay in Botswana.
CONTROL	Botswana requires a local control-plane slice. Only signed policy, claims and non-sensitive deployment metadata may enter, subject to legal approval.
SERVICES	Cells repeat the same logical service model. Web assets, the MFE registry, replicas, workers and adapters are served inside the placement boundary.
TEMPORAL	Workflow hosting follows placement. The runtime, workers and persistence are cell-resident; Botswana is self-hosted in-country.
STORAGE	Every stateful technology follows placement. Service databases and evidence repositories are independently backed up and recovered inside the permitted geography.
SECURITY	Controls apply at every ingress and hop. TLS, encryption at rest, RBAC/ABAC, short-lived credentials, masking and immutable audit are cell-local obligations.
VALIDATE	This is a proposed compliance posture. Botswana legal, cloud availability, network, support and recovery constraints still require formal validation.

KEY · SHEET 2

Green boundary — trust and policy authority	Blue boundary — edge and entry: country endpoints, routing	Cyan boundary — shared deployment cell	Amber boundary — sovereign cell; restricted by law
Tile — a deployed element with a stated home	Amber-filled tile — policy or trust artefact	Solid blue line — synchronous request/reply (HTTPS, NATS RPC)	Solid green line — signed policy, claims and placement intent
Dotted cyan line — NATS gateway; approved accounts and subjects only	Dashed red line — prohibited path; drawn to show it is barred only	Nesting — an element inside a cell runs in that cell	Arrowhead — every line is one-way; it points at the receiver

ACRONYMS

ABAC Attribute-based access control	NATS The messaging system; a product name, not an abbreviation
API Application programming interface	RBAC Role-based access control
AU Australia (ISO country code)	RPO Recovery point objective: tolerated data loss
AUD Australian dollar (ISO currency code)	RTO Recovery time objective: tolerated downtime
BW Botswana (ISO country code)	S3 The object-storage API; S3-compatible means any vendor implementing it
BWP Botswana pula (ISO currency code)	TLS Transport Layer Security
HA High availability	WAF Web application firewall
JS JetStream, the NATS persistence layer (not JavaScript)	ZA South Africa (ISO country code)
JWT JSON Web Token	ZAR South African rand (ISO currency code)
KV Key/value store (NATS KV)	
MFE Micro-frontend	